

2. B) span: $\{I, X, Z, XZ\}$ is correctable (Show).

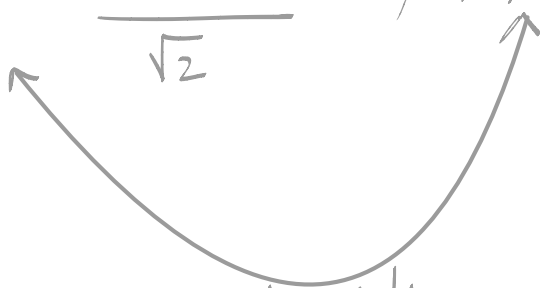
Let's assume V is a correctable error. Then the KL conditions should hold:

$$P E_a^\dagger E_b P = \alpha_{ab} P$$

$$\alpha_{ab} = \langle 0_L | E_a^\dagger E_b | 0_L \rangle = \langle 1_L | E_a^\dagger E_b | 1_L \rangle$$

$$0 = \langle 0_L | E_a^\dagger E_b | 1_L \rangle = \langle 1_L | E_a^\dagger E_b | 0_L \rangle$$

$$|0_L\rangle = \frac{|0\rangle + |4\rangle}{\sqrt{2}} ; |1_L\rangle = \frac{|2\rangle + |6\rangle}{\sqrt{2}}$$



codewords.

(i) $E_a = I, E_b = X; E_a^\dagger E_b = X$

$$\langle 0_L | X | 0_L \rangle = \left(\frac{\langle 0| + \langle 4|}{\sqrt{2}} \right) \left(\frac{|1\rangle + |5\rangle}{\sqrt{2}} \right)$$

$$= 0$$

$$\langle 1_L | X | 1_L \rangle = \left(\frac{\langle 2 | + \langle 6 |}{\sqrt{2}} \right) \left(\frac{|3\rangle + |7\rangle}{\sqrt{2}} \right) = 0$$

$$\langle 0_L | X | 1_L \rangle = \left(\frac{\langle 0 | + \langle 4 |}{\sqrt{2}} \right) \left(\frac{|3\rangle + |7\rangle}{\sqrt{2}} \right) = 0.$$

ii. $E_a^\dagger = X; E_b = Z; E_a^\dagger E_b = X^\dagger Z$

$$\langle 0_L | X^\dagger Z | 0_L \rangle = \left(\frac{\langle 1 | + \langle 5 |}{\sqrt{2}} \right) \left(\frac{|0\rangle - |4\rangle}{\sqrt{2}} \right)$$

$$\langle 1_L | X^\dagger Z | 1_L \rangle = \left(\frac{\langle 7 | + \langle 3 |}{\sqrt{2}} \right) \left(\frac{|2\rangle - |6\rangle}{\sqrt{2}} \right) i$$

$$\langle 0_L | X^\dagger Z | 1_L \rangle = \left(\frac{\langle 1 | + \langle 5 |}{\sqrt{2}} \right) \left(\frac{|2\rangle - |6\rangle}{\sqrt{2}} \right) i = 0$$

iii. $E_a^\dagger = Z; E_b = XZ; E_a^\dagger E_b = Z^\dagger XZ$

$$= Z^\dagger \cdot Z X e^{-i\pi/4}$$

$$= X e^{-i\pi/4}$$

$$\langle 0_L | X e^{-i\pi/4} | 0_L \rangle = e^{-i\pi/4} \left(\frac{\langle 0 | + \langle 4 |}{\sqrt{2}} \right) \left(\frac{|1\rangle + |5\rangle}{\sqrt{2}} \right) = 0$$

$$\langle 1_L | X e^{-i\pi/4} | 1_L \rangle = e^{-i\pi/4} \left(\frac{\langle 2| + \langle 6|}{\sqrt{2}} \right) \left(\frac{|3\rangle + |7\rangle}{\sqrt{2}} \right) = 0.$$

$$\langle 0_L | X e^{-i\pi/4} | 1_L \rangle = e^{-i\pi/4} \left(\frac{\langle 0| + \langle 4|}{\sqrt{2}} \right) \left(\frac{|3\rangle + |7\rangle}{\sqrt{2}} \right) = 0.$$

$$\text{iv. } \langle 0_L | XZ | 0_L \rangle = \left(\frac{\langle 0| + \langle 4|}{\sqrt{2}} \right) \left(\frac{|1\rangle - |5\rangle}{\sqrt{2}} \right) = 0$$

$$\langle 1_L | XZ | 1_L \rangle = i \left(\frac{\langle 2| + \langle 6|}{\sqrt{2}} \right) \left(\frac{|3\rangle - |7\rangle}{\sqrt{2}} \right) = 0$$

$$\langle 0_L | XZ | 1_L \rangle = i \left(\frac{\langle 0| + \langle 4|}{\sqrt{2}} \right) \left(\frac{|3\rangle - |7\rangle}{\sqrt{2}} \right) = 0$$

All the other combinations will follow symmetry.

Is this choice unique? Why or why not?

No. Consider a set: $\{ I, XZ^4, X^4Z, XZ \}$

$\uparrow \quad \quad \uparrow$
 $S_2 \quad S_1$

This is still correctable but the error set is different. So, the original error set is NOT unique.